

GDP and Emissions Database

Version 1.0

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Data is documented in the following table structure:

Variable	Range	Description
Name in Dataset	FirstObs–LastObs	Title; Unit Construction, Source

The datasets are provided both as individual `.csv` files and as a consolidated `.xlsx` file. The `.csv` files are titled `cc_data.csv`, where `cc` is the two-letter country code in lowercase. Each `.csv` file uses a comma as the value separator, a dot as the decimal separator, and the first row as the header. The consolidated `.xlsx` file contains the data organized in Excel sheets titled `CC`, where `CC` is the two-letter country code in uppercase.

Contents

1. Australia (AU)	3
2. Austria (AT)	5
3. Belgium (BE)	7
4. Canada (CA)	9
5. Denmark (DK)	11

6. Finland (FI)	13
7. France (FR)	15
8. Germany (DE)	16
9. Greece (GR)	18
10. Ireland (IE)	20
11. Italy (IT)	22
12. Japan (JP)	24
13. Luxembourg (LU)	26
14. Netherlands (NL)	28
15. New Zealand (NZ)	30
16. Norway (NO)	32
17. Portugal (PT)	34
18. Spain (ES)	36
19. Sweden (SE)	38
20. Switzerland (CH)	40
21. United Kingdom (GB)	42
22. United States (US)	43
A. Construction of Employment- and Hour-Based TFP Indices	45

1. Australia (AU)

Variable	Range	Description
gdp	1950–2023	Real GDP; 2017 AUD, millions The following series rescaled to match the value of the series Australian Bureau of Statistics (2026<i>e</i>) in 2017: i) Since 1959: Australian Bureau of Statistics (2026<i>d</i>) ii) Prior to 1959: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2023	Real net capital stock; 2017 AUD, millions The following series rescaled to match the value of the series Australian Bureau of Statistics (2026<i>c</i>) in 2017: i) Since 1959: Australian Bureau of Statistics (2026<i>b</i>) ii) Prior to 1959: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1994: OECD (2026<i>b</i>) ii) Prior to 1994: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions i) Since 1994: OECD (2026<i>c</i>) ii) Prior to 1994: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1860–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026<i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1959: Share of Australian Bureau of Statistics (2026<i>a</i>) in the total of Australian Bureau of Statistics (2026<i>a</i>) and Australian Bureau of Statistics (2026<i>f</i>) ii) Prior to 1959: Feenstra et al. (2015, LABSH)
tfpe	1950–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

2. Austria (AT)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026<i>d</i>) in 2017: i) Since 1995: Eurostat (2026<i>c</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026<i>h</i>) in 2017: i) Since 1995: Eurostat (2026<i>g</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026<i>b</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026<i>f</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1819–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026<i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026<i>a</i>) in the total of Eurostat (2026<i>a</i>) and Eurostat (2026<i>e</i>) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

3. Belgium (BE)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026<i>d</i>) in 2017: i) Since 1995: Eurostat (2026<i>c</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026<i>h</i>) in 2017: i) Since 1995: Eurostat (2026<i>g</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026<i>b</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026<i>f</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1830–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026<i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026<i>a</i>) in the total of Eurostat (2026<i>a</i>) and Eurostat (2026<i>e</i>) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

4. Canada (CA)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 CAD, millions The following series rescaled to match the value of the series Statistics Canada (2026<i>e</i>) in 2017: i) Since 1981: Statistics Canada (2026<i>d</i>) ii) Prior to 1981: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPA)
nks	1950–2024	Real net capital stock; 2017 CAD, millions The following series rescaled to match the end of year value of the series Statistics Canada (2026<i>e</i>) in 2017: i) Since 2009: End of year values of Statistics Canada (2026<i>b</i>) ii) Prior to 2009: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1970: OECD (2026<i>b</i>) ii) Prior to 1970: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions i) Since 1997: OECD (2026<i>c</i>) ii) Prior to 1997: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1785–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026<i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1981: Share of Statistics Canada (2026<i>a</i>) in the total of Statistics Canada (2026<i>a</i>) and Statistics Canada (2026<i>f</i>) ii) Prior to 1981: Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

5. Denmark (DK)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 DKK, millions The following series rescaled to match the value of the series Eurostat (2026<i>d</i>) in 2017: i) Since 1975: Eurostat (2026<i>c</i>) ii) Prior to 1975: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 DKK, millions The following series rescaled to match the end of year value of the series Eurostat (2026<i>h</i>) in 2017: i) Since 1975: Eurostat (2026<i>g</i>) ii) Prior to 1975: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1975: Eurostat (2026<i>b</i>) ii) Prior to 1975: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1975: Eurostat (2026<i>f</i>) ii) Prior to 1975: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1843–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026<i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026<i>a</i>) in the total of Eurostat (2026<i>a</i>) and Eurostat (2026<i>e</i>) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

6. Finland (FI)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026<i>d</i>) in 2017: i) Since 1980: Eurostat (2026<i>c</i>) ii) Prior to 1980: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026<i>h</i>) in 2017: i) Since 1980: Eurostat (2026<i>g</i>) ii) Prior to 1980: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1980: Eurostat (2026<i>b</i>) ii) Prior to 1980: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1980: Eurostat (2026<i>f</i>) ii) Prior to 1980: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1860–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026<i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1975: Share of Eurostat (2026<i>a</i>) in the total of Eurostat (2026<i>a</i>) and Eurostat (2026<i>e</i>) ii) Prior to 1975: Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

7. France (FR)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026<i>d</i>) in 2017: i) Since 1975: Eurostat (2026<i>c</i>) ii) Prior to 1975: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026<i>h</i>) in 2017: i) Since 1978: Eurostat (2026<i>g</i>) ii) Prior to 1978: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1978: Eurostat (2026<i>b</i>) ii) Prior to 1978: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1978: Eurostat (2026<i>f</i>) ii) Prior to 1978: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1810–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026 <i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1949–2024	Labor share Share of Eurostat (2026 <i>a</i>) in the total of Eurostat (2026 <i>a</i>) and Eurostat (2026 <i>e</i>)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

8. Germany (DE)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026 <i>d</i>) in 2017: i) Since 1991: Eurostat (2026 <i>c</i>) ii) Prior to 1991: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)

Variable	Range	Description
nks	1950–2024	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026<i>h</i>) in 2017: i) Since 1995: Eurostat (2026<i>g</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026<i>b</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026<i>f</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1945–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026<i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026<i>a</i>) in the total of Eurostat (2026<i>a</i>) and Eurostat (2026<i>e</i>)

Variable	Range	Description
		ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

9. Greece (GR)

Variable	Range	Description
gdp	1951–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026 <i>d</i>) in 2017: i) Since 1995: Eurostat (2026 <i>c</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1951–2023	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026 <i>h</i>) in 2017: i) Since 1995: Eurostat (2026 <i>g</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)

Variable	Range	Description
emp	1951–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026<i>b</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1951–2024	Hours worked; millions The following series divided by 1000:s i) Since 1995: Eurostat (2026<i>f</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1913–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026<i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1951–2024	Labor share i) Since 1995: Share of Eurostat (2026<i>a</i>) in the total of Eurostat (2026<i>a</i>) and Eurostat (2026<i>e</i>) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1951–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017

Variable	Range	Description
tfph	1951–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1951–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

10. Ireland (IE)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026 <i>d</i>) in 2017: i) Since 1995: Eurostat (2026 <i>c</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2023	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026 <i>h</i>) in 2017: i) Since 1995: Eurostat (2026 <i>g</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026 <i>b</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)

Variable	Range	Description
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026<i>f</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1924–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026<i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026<i>a</i>) in the total of Eurostat (2026<i>a</i>) and Eurostat (2026<i>e</i>) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026)

Variable	Range	Description
		ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

11. Italy (IT)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026 <i>d</i>) in 2017: i) Since 1995: Eurostat (2026 <i>c</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026 <i>h</i>) in 2017: i) Since 1995: Eurostat (2026 <i>g</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026 <i>b</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026 <i>f</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)

Variable	Range	Description
co2	1860–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026 <i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026 <i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026 <i>a</i>) in the total of Eurostat (2026 <i>a</i>) and Eurostat (2026 <i>e</i>) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

12. Japan (JP)

Variable	Range	Description
gdp	1950–2023	Real GDP; 2017 JPY, millions The following series multiplied by 1000 after being rescaled to match the value of the series Statistics Bureau of Japan (2026<i>c</i>) in 2017: i) Since 1994: Statistics Bureau of Japan (2026<i>b</i>) ii) Prior to 1994: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2023	Real net capital stock; 2017 JPY, millions The following series multiplied by 1000 after being rescaled to match the end of year value of the series Statistics Bureau of Japan (2026<i>f</i>) in 2017: i) Since 1994: Statistics Bureau of Japan (2026<i>e</i>) ii) Prior to 1994: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1970: OECD (2026<i>b</i>) ii) Prior to 1970: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions i) Since 1970: OECD (2026<i>c</i>) ii) Prior to 1970: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1950–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026<i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2023	Labor share i) Since 1994: Share of Statistics Bureau of Japan (2026<i>a</i>) in the total of Statistics Bureau of Japan (2026<i>a</i>) and Statistics Bureau of Japan (2026<i>d</i>) ii) Prior to 1994: Feenstra et al. (2015, LABSH)
tfpe	1950–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

13. Luxembourg (LU)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026<i>d</i>) in 2017: i) Since 1995: Eurostat (2026<i>c</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2023	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026<i>h</i>) in 2017: i) Since 1995: Eurostat (2026<i>g</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026<i>b</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1970–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026<i>f</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1945–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026<i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026<i>a</i>) in the total of Eurostat (2026<i>a</i>) and Eurostat (2026<i>e</i>) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1970–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

14. Netherlands (NL)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026<i>d</i>) in 2017: i) Since 1995: Eurostat (2026<i>c</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026<i>h</i>) in 2017: i) Since 1995: Eurostat (2026<i>g</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026<i>b</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1970–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026<i>f</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1846–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026<i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026<i>a</i>) in the total of Eurostat (2026<i>a</i>) and Eurostat (2026<i>e</i>) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1970–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

15. New Zealand (NZ)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 NZD, millions The following series rescaled to match the value of the series Statistics New Zealand (2026<i>c</i>) in 2017: i) Since 1987: Statistics New Zealand (2026<i>b</i>) ii) Prior to 1987: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 NZD, millions The following series rescaled to match the end of year value of the series Statistics New Zealand (2026<i>f</i>) in 2017: i) Since 1971: Statistics New Zealand (2026<i>e</i>) ii) Prior to 1971: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1989: OECD (2026<i>b</i>) ii) Prior to 1989: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1970–2024	Hours worked; millions i) Since 1971: OECD (2026<i>c</i>) ii) Prior to 1971: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1878–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026<i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2023	Labor share i) Since 1971: Share of Statistics New Zealand (2026<i>a</i>) in the total of Statistics New Zealand (2026<i>a</i>) and Statistics New Zealand (2026<i>d</i>) ii) Prior to 1971: i) backdated using growth rates (previous period) of Feenstra et al. (2015, LABSH)
tfpe	1950–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1970–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

16. Norway (NO)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 NOK, millions The following series rescaled to match the value of the series Statistics Norway (2026<i>d</i>) in 2017: i) Since 1970: Statistics Norway (2026<i>c</i>) ii) Prior to 1970: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 NOK, millions The following series rescaled to match the end of year value of the series Statistics Norway (2026<i>f</i>) in 2017: i) Since 1970: Statistics Norway (2026<i>e</i>) ii) Prior to 1970: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1970: OECD (2026<i>b</i>) ii) Prior to 1970: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions i) Since 1970: OECD (2026<i>c</i>) ii) Prior to 1970: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1835–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026<i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2023	Labor share i) Since 1970: Share of Statistics Norway (2026<i>a</i>) in the total of Statistics Norway (2026<i>a</i>), Statistics Norway (2026<i>g</i>) and Statistics Norway (2026<i>b</i>) ii) Prior to 1970: Feenstra et al. (2015, LABSH)
tfpe	1950–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

17. Portugal (PT)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026<i>d</i>) in 2017: i) Since 1995: Eurostat (2026<i>c</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2023	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026<i>h</i>) in 2017: i) Since 2000: Eurostat (2026<i>g</i>) ii) Prior to 2000: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026<i>b</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026<i>f</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1870–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026<i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026<i>a</i>) in the total of Eurostat (2026<i>a</i>) and Eurostat (2026<i>e</i>) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

18. Spain (ES)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026<i>d</i>) in 2017: i) Since 1995: Eurostat (2026<i>c</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2023	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026<i>h</i>) in 2017: i) Since 1995: Eurostat (2026<i>g</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026<i>b</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026<i>f</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1830–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026<i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026<i>a</i>) in the total of Eurostat (2026<i>a</i>) and Eurostat (2026<i>e</i>) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

19. Sweden (SE)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 SEK, millions The following series rescaled to match the value of the series Eurostat (2026<i>d</i>) in 2017: i) Since 1993: Eurostat (2026<i>c</i>) ii) Prior to 1993: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2022	Real net capital stock; 2017 SEK, millions The following series rescaled to match the end of year value of the series Eurostat (2026<i>h</i>) in 2017: i) Since 1993: Eurostat (2026<i>g</i>) ii) Prior to 1993: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1993: Eurostat (2026<i>b</i>) ii) Prior to 1993: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1993: Eurostat (2026<i>f</i>) ii) Prior to 1993: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1839–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026<i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026<i>a</i>) in the total of Eurostat (2026<i>a</i>) and Eurostat (2026<i>e</i>) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2022	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2022	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

20. Switzerland (CH)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 CHF, millions The following series rescaled to match the value of the series Bundesamt für Statistik (2026<i>e</i>) in 2017: i) Since 1996: Cumulative product of one plus the series Bundesamt für Statistik (2026<i>d</i>) divided by 100 ii) Prior to 1996: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 CHF, millions The following series rescaled to match the end of year value of the series Bundesamt für Statistik (2026<i>g</i>) in 2017: i) Since 1996: Bundesamt für Statistik (2026<i>f</i>) ii) Prior to 1996: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands The following series divided by 1000: i) Since 2010: Bundesamt für Statistik (2026<i>c</i>) ii) Prior to 2010: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000000: i) Since 2010: Bundesamt für Statistik (2026<i>i</i>) ii) Prior to 2010: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1858–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026<i>a</i>)

Variable	Range	Description
		ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026 <i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Bundesamt für Statistik (2026 <i>a</i>) in the total of Bundesamt für Statistik (2026 <i>a</i>), Bundesamt für Statistik (2026 <i>h</i>) and Bundesamt für Statistik (2026 <i>b</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

21. United Kingdom (GB)

Variable	Range	Description
gdp	1948–2024	Real GDP; 2017 GBP, millions The following series rescaled to match the value of the series Office for National Statistics (2026 <i>c</i>) in 2017: Office for National Statistics (2026 <i>b</i>)
nks	1950–2024	Real net capital stock; 2017 GBP, millions The following series multiplied by 1000 after being rescaled to match the end of year value of the series Office for National Statistics (2026 <i>f</i>) in 2017: i) Since 1995: Office for National Statistics (2026 <i>e</i>) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1980: OECD (2026 <i>b</i>) ii) Prior to 1980: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions i) Since 1980: OECD (2026 <i>c</i>) ii) Prior to 1980: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1751–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026 <i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026 <i>d</i>)

Variable	Range	Description
		ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1948–2024	Labor share Share of Office for National Statistics (2026 <i>a</i>) in the total of Office for National Statistics (2026 <i>a</i>) and Office for National Statistics (2026 <i>d</i>)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

22. United States (US)

Variable	Range	Description
gdp	1929–2024	Real GDP; 2017 USD, millions The following series rescaled to match the value of the series Bureau of Economic Analysis (2026 <i>b</i>) in 2017: Bureau of Economic Analysis (2026 <i>a</i>)
nks	1925–2024	Real net capital stock; 2017 USD, millions The following series rescaled to match the end of year value of the series Bureau of Economic Analysis (2026 <i>d</i>) in 2017:

Variable	Range	Description
		Bureau of Economic Analysis (2026 <i>c</i>)
emp	1948–2024	Employment; jobs, thousands Bureau of Labor Statistics (2026 <i>a</i>) multiplied by 1000
hrs	1948–2024	Hours worked; millions Bureau of Labor Statistics (2026 <i>b</i>) multiplied by 1000
co2	1800–2022	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026 <i>a</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026 <i>d</i>) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1948–2024	Labor share Bureau of Labor Statistics (2026 <i>c</i>)
tfpe	1948–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1948–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

A. Construction of Employment- and Hour-Based TFP Indices

The employment- and hour-based TFP indices tfpe and tfph , respectively, are calculated based on gdp and a chained Törnqvist index of combined inputs in line with the TFP calculation of the Bureau of Labor Statistics (<https://www.bls.gov/opub/hom/opt/calculation.htm>) and the methodological note to experimental statistics on (crude) multifactor productivity on the corresponding eurostat webpage (<https://ec.europa.eu/eurostat/web/experimental-statistics/multifactor-productivity>). In short, given observations for $t = 1, 2, \dots$, tfpe and tfph (prior to rescaling to match the value 1 in 2017) are calculated as follows:¹

$$\begin{aligned}\text{tfpe}_t &:= \frac{\text{gdp}_t}{\prod_{s=2}^t \exp((1 - w_{L,s})\Delta \ln(\text{nks}_s) + w_{L,s}\Delta \ln(\text{emp}_s))}, \\ \text{tfph}_t &:= \frac{\text{gdp}_t}{\prod_{s=2}^t \exp((1 - w_{L,s})\Delta \ln(\text{nks}_s) + w_{L,s}\Delta \ln(\text{hrs}_s))}, \\ w_{L,t} &:= \frac{1}{2}(\text{ls}_{t-1} + \text{ls}_t),\end{aligned}$$

where the variables gdp , nks , emp , hrs , and ls are as described in the corresponding sections above.

¹Note that empty products, i. e., here the denominators of tfpe_t and tfph_t for $t = 1$, evaluate to one.

References

- Australian Bureau of Statistics (2026*a*), ‘Compensation of Employees. Current Prices. AUD, Millions. Annual’. URL: <https://www.abs.gov.au/statistics/economy/national-accounts/australian-system-national-accounts/latest-release>. Table: 5204.0/06, Series ID: A2421040T.
- Australian Bureau of Statistics (2026*b*), ‘End-Year Net Capital Stock. Chain Volume Measures. AUD, Millions. Annual’. URL: <https://www.abs.gov.au/statistics/economy/national-accounts/australian-national-accounts-national-income-expenditure-and-product/latest-release>. Table: 5204.0/56, Series ID: A2422574C.
- Australian Bureau of Statistics (2026*c*), ‘End-Year Net Capital Stock. Current Prices. AUD, Millions. Annual’. URL: <https://www.abs.gov.au/statistics/economy/national-accounts/australian-national-accounts-national-income-expenditure-and-product/latest-release>. Table: 5204.0/56, Series ID: A2422571W.
- Australian Bureau of Statistics (2026*d*), ‘Gross Domestic Product. Chain Volume Measures. AUD, Millions. Annual’. URL: <https://www.abs.gov.au/statistics/economy/national-accounts/australian-national-accounts-national-income-expenditure-and-product/latest-release>. Table: 5204.0/2, Series ID: A2420912W.
- Australian Bureau of Statistics (2026*e*), ‘Gross Domestic Product. Current Prices. AUD, Millions. Annual’. URL: <https://www.abs.gov.au/statistics/economy/national-accounts/australian-national-accounts-national-income-expenditure-and-product/latest-release>. Table: 5204.0/2. Series ID: A2420638T.
- Australian Bureau of Statistics (2026*f*), ‘Gross Operating Surplus. Current Prices. AUD, Millions. Annual’. URL: <https://www.abs.gov.au/statistics/economy/national-accounts/australian-system-national-accounts/latest-release>. Table: 5204.0/06, Series ID: A2421313L.
- Bundesamt für Statistik (2026*a*), ‘Compensation of Employees. Current Prices, Millions. Annual’. URL: <https://www.bfs.admin.ch/bfs/en/home/statistics/catalogue.assetdetail.36182089.html>. CLASSIFICATION: D1, UNIT_MEAS: MCHF.
- Bundesamt für Statistik (2026*b*), ‘Consumption of Fixed Capital. Current Prices, Millions. Annual’. URL: <https://www.bfs.admin.ch/bfs/en/home/statistics/catalogue.assetdetail.36182089.html>. CLASSIFICATION: P51C, UNIT_MEAS: MCHF.

- Bundesamt für Statistik (2026*c*), ‘Employment. Persons. Annual’. URL: <https://www.bfs.admin.ch/bfs/de/home/statistiken/katalog.assetdetail.36190829.html>. INDICATORS_DE: HERKUNFT, GENDER_DE: Total, NAT_DE: Total, DETAILS_DE: Total, FREQ: Y.
- Bundesamt für Statistik (2026*d*), ‘Gross Domestic Product. Annual Percentage Change in Chain Volume Measure. CHF, Millions. Annual’. URL: <https://www.bfs.admin.ch/bfs/en/home/statistics/catalogue.assetdetail.36182125.html>. NOGA: B1GQ, VARIABLE: GVA, UNIT_MEAS: ACPP.
- Bundesamt für Statistik (2026*e*), ‘Gross Domestic Product. Current Prices. CHF, Millions. Annual’. URL: <https://www.bfs.admin.ch/bfs/en/home/statistics/catalogue.assetdetail.36182125.html>. NOGA: B1GQ, VARIABLE: GVA, UNIT_MEAS: MCHF.
- Bundesamt für Statistik (2026*f*), ‘Net Non Financial Capital Stock. Chain Volume Measure. CHF, Millions. Annual’. URL: <https://www.bfs.admin.ch/bfs/en/home/statistics/catalogue.assetdetail.36181954.html>. CLASSIFICATION: TOTAL, UNIT_MEAS: MCHFC.
- Bundesamt für Statistik (2026*g*), ‘Net Non Financial Capital Stock. Current Prices. CHF, Millions. Annual’. URL: <https://www.bfs.admin.ch/bfs/en/home/statistics/catalogue.assetdetail.36181954.html>. CLASSIFICATION: TOTAL, UNIT_MEAS: MCHF.
- Bundesamt für Statistik (2026*h*), ‘Net Operating Surplus and Mixed Income. Current Prices, Millions. Annual’. URL: <https://www.bfs.admin.ch/bfs/en/home/statistics/catalogue.assetdetail.36182089.html>. CLASSIFICATION: B2_B3N, UNIT_MEAS: MCHF.
- Bundesamt für Statistik (2026*i*), ‘Total Hours Worked. Annual’. URL: <https://www.bfs.admin.ch/bfs/de/home/statistiken/katalog.assetdetail.35227378.html>. SEX: _T, NAT: _T, WORKTIME: _T, NOGA1: _T, SECTOR: _T, EMP_STATUS: _T, REGION: _T.
- Bureau of Economic Analysis (2026*a*), ‘Gross Domestic Product. Chain Volume Measure. USD, Millions. Annual’. URL: <https://www.bea.gov/itable/national-gdp-and-personal-income>. Table 1.1.6. Real Gross Domestic Product, Chained Dollars. Time series key: A191RX.
- Bureau of Economic Analysis (2026*b*), ‘Gross Domestic Product. Current Prices. USD, Millions. Annual’. URL: <https://www.bea.gov/itable/national-gdp-and-personal-income>. Table 1.1.5. Gross Domestic Product. Time series key: A191RC.
- Bureau of Economic Analysis (2026*c*), ‘Net Stock of Fixed Assets. Chain Volume Measure. USD, Millions. Annual’. URL: <https://www.bea.gov/itable/fixed-assets>. Table 1.2. Chain-Type Quantity Indexes for Net Stock of Fixed Assets and Consumer Durable Goods. Time series key: kcttotlles00.

- Bureau of Economic Analysis (2026*d*), ‘Net Stock of Fixed Assets. Current Prices. USD, Millions. Annual’. URL: <https://www.bea.gov/itable/fixed-assets>. Table 1.1. Current-Cost Net Stock of Fixed Assets and Consumer Durable Goods. Time series key: k1ttotlles00.
- Bureau of Labor Statistics (2026*a*), ‘Employment. Jobs, Millions. Annual’. URL: <https://www.bls.gov/productivity/tables>. Historical total factor productivity measures (SIC 1948-87 linked to NAICS 1987-present). Private business sector. Employment.
- Bureau of Labor Statistics (2026*b*), ‘Hours Worked. Billions. Annual’. URL: <https://www.bls.gov/productivity/tables>. Historical total factor productivity measures (SIC 1948-87 linked to NAICS 1987-present). Private business sector. Hours worked.
- Bureau of Labor Statistics (2026*c*), ‘Labor Share. Annual’. URL: <https://www.bls.gov/productivity/tables>. Historical total factor productivity measures (SIC 1948-87 linked to NAICS 1987-present). Private business sector. Labor share.
- Eurostat (2026*a*), ‘Compensation of Employees. Current Prices. National Currency, Millions. Annual’. URL: https://doi.org/10.2908/NASA_10_NF_TR. Time series key: NASA_10_NF_TR.A.CP_MNAC.RECV.D1.S1.CC, where CC is the two-digit country code in uppercase letters (EL for Greece).
- Eurostat (2026*b*), ‘Employment. Persons, Thousands. Annual’. URL: https://doi.org/10.2908/NAMA_10_A64_E. Time series key: NAMA_10_A64_E.A.THS_PER.TOTAL.EMP_DC.CC, where CC is the two-digit country code in uppercase letters (EL for Greece).
- Eurostat (2026*c*), ‘Gross Domestic Product. Chain Volume Measure. National Currency, Millions. Annual’. URL: https://doi.org/10.2908/NAMA_10_GDP. Time series key: NAMA_10_GDP.A.CLV20_MNAC.B1GQ.CC, where CC is the two-digit country code in uppercase letters (EL for Greece).
- Eurostat (2026*d*), ‘Gross Domestic Product. Current Prices. National Currency, Millions. Annual’. URL: https://doi.org/10.2908/NAMA_10_GDP. Time series key: NAMA_10_GDP.A.CP_MNAC.B1GQ.CC, where CC is the two-digit country code in uppercase letters (EL for Greece).
- Eurostat (2026*e*), ‘Gross Operating Surplus. Current Prices. National Currency, Millions. Annual’. URL: https://doi.org/10.2908/NASA_10_NF_TR. Time series key: NASA_10_NF_TR.A.CP_MNAC.RECV.B2G.S1.CC, where CC is the two-digit country code in uppercase letters (EL for Greece).
- Eurostat (2026*f*), ‘Hours Worked. Thousands. Annual’. URL: https://doi.org/10.2908/NAMA_10_A64_E. Time series key: NAMA_10_A64_E.A.THS_HW.TOTAL.EMP_DC.CC, where CC is the two-digit country code in uppercase letters (EL for Greece).

- Eurostat (2026g), ‘Total Fixed Assets (Net). Chain Volume Measure. National Currency, Millions. Annual’. URL: https://doi.org/10.2908/NAMA_10_NFA_ST. Time series key: NAMA_10_NFA_ST.A.CLV20_MNAC.TOTAL.N11N.CC, where CC is the two-digit country code in uppercase letters (EL for Greece).
- Eurostat (2026h), ‘Total Fixed Assets (Net). Current Prices. National Currency, Millions. Annual’. URL: https://doi.org/10.2908/NAMA_10_NFA_ST. Time series key: NAMA_10_NFA_ST.A.CRC_MNAC.TOTAL.N11N.CC, where CC is the two-digit country code in uppercase letters (EL for Greece).
- Feenstra, R., Inklarr, R. and Timmer, M. (2015), ‘The Next Generation of the Penn World Table’, *American Economic Review* **105**, 3150–3182. Data obtained from Penn World Table version 11.0, URL: <https://www.rug.nl/ggdc/productivity/pwt/>.
- Hefner, M. and Marland, G. (2025), ‘Global, Regional, and National Fossil-Fuel CO2 Emissions: 1751–2021 CDIAC-FF’. Research Institute for Environment, Energy, and Economics, Appalachian State University. URL: https://rieee.appstate.edu/projects-programs/cdiac.nation.1751_2021.xlsx, Total CO2 emissions from fossil-fuels and cement production (thousand metric tons of C).
- OECD (2026a), ‘Carbon Dioxide Emissions. Metric Tons, Thousands. Annual’. URL: [https://data-explorer.oecd.org/vis?df\[ds\]=dsDisseminateFinalDMZ&df\[id\]=DSD_AIR_GHG%40DF_AIR_GHG&df\[ag\]=OECD.ENV.EPI](https://data-explorer.oecd.org/vis?df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_AIR_GHG%40DF_AIR_GHG&df[ag]=OECD.ENV.EPI). Time series key: CCC.A.CO2._T.T_CO2E, where CCC is the three-digit country code in uppercase letters.
- OECD (2026b), ‘Employment. Persons, Thousands. Annual’. URL: [https://data-explorer.oecd.org/vis?df\[ds\]=dsDisseminateFinalDMZ&df\[id\]=DSD_PDB%40DF_PDB&df\[ag\]=OECD.SDD.TPS](https://data-explorer.oecd.org/vis?df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_PDB%40DF_PDB&df[ag]=OECD.SDD.TPS). Time series key: CCC.A.EMP._T...N., where CCC is the three-digit country code in uppercase letters.
- OECD (2026c), ‘Hours Worked for Total Employment. Millions. Annual’. URL: [https://data-explorer.oecd.org/vis?df\[ds\]=dsDisseminateFinalDMZ&df\[id\]=DSD_PDB%40DF_PDB&df\[ag\]=OECD.SDD.TPS](https://data-explorer.oecd.org/vis?df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_PDB%40DF_PDB&df[ag]=OECD.SDD.TPS). Time series key: CCC.A.HRSTO._T...N., where CCC is the three-digit country code in uppercase letters.
- OECD (2026d), ‘Sulfur Oxides Emissions. Metric Tons, Thousands. Annual’. URL: [https://data-explorer.oecd.org/vis?df\[ds\]=DisseminateFinalDMZ&df\[id\]=DSD_AIR_EMISSIONS%40DF_AIR_EMISSIONS&df\[ag\]=OECD.ENV.EPI](https://data-explorer.oecd.org/vis?df[ds]=DisseminateFinalDMZ&df[id]=DSD_AIR_EMISSIONS%40DF_AIR_EMISSIONS&df[ag]=OECD.ENV.EPI). Time series key: CCC.A.SOX.T_EM_MM.T, where CCC is the three-digit country code in uppercase letters.
- Office for National Statistics (2026a), ‘Compensation of Employees. Current Prices. GBP, Millions. Annual’. URL: <https://www.ons.gov.uk/economy/grossdomesticproductgdp/datasets/bluebook>. Time series key: HAEA.

- Office for National Statistics (2026*b*), ‘Gross Domestic Product. Chain Volume Measure. GBP, Millions. Annual’. URL: <https://www.ons.gov.uk/economy/grossdomesticproductgdp/datasets/bluebook>. Time series key: ABMI.
- Office for National Statistics (2026*c*), ‘Gross Domestic Product. Current Prices. GBP, Millions. Annual’. URL: <https://www.ons.gov.uk/economy/grossdomesticproductgdp/datasets/bluebook>. Time series key: YBHA.
- Office for National Statistics (2026*d*), ‘Gross Operating Surplus. Current Prices. GBP, Millions. Annual’. URL: <https://www.ons.gov.uk/economy/grossdomesticproductgdp/datasets/bluebook>. Time series key: ABNF.
- Office for National Statistics (2026*e*), ‘Net Capital Stock. Chain Volume Measure. GBP, Millions. Annual’. URL: <https://www.ons.gov.uk/economy/nationalaccounts/uksectoraccounts/datasets/capitalstock>. Time series key: MLR3.
- Office for National Statistics (2026*f*), ‘Net Capital Stock. Current Prices. GBP, Millions. Annual’. URL: <https://www.ons.gov.uk/economy/nationalaccounts/uksectoraccounts/datasets/capitalstock>. Time series key: MJ12.
- Smith, S., van Aardenne, J., Klimont, Z., Andres, R., Volke, A. and Delgado Arias, S. (2011), ‘Anthropogenic Sulfur Dioxide Emissions: 1850–2005’, *Atmospheric Chemistry and Physics* **11**, 1101–1116. National and Regional Data Set by Source Category, Version 2.86 (Version 2.86).
- Statistics Bureau of Japan (2026*a*), ‘Compensation of Employees. Current Prices. JPY, Billions. Annual’. URL: <https://www.e-stat.go.jp/en/stat-search/files?tclass=000001225282&cycle=7&year=20230>. Income and Outlay Accounts classified by Institutional Sectors. Total Economy. Allocation of Primary Income Account. CY. Compensation of Employees, receivable.
- Statistics Bureau of Japan (2026*b*), ‘Gross Domestic Product. Chain Volume Measure. JPY, Billions. Annual’. URL: <https://www.e-stat.go.jp/en/stat-search/files?tclass=000001225282&cycle=7&year=20230>. Gross Domestic Product classified by Economic Activities. In real terms. Calendar Year. Gross domestic product.
- Statistics Bureau of Japan (2026*c*), ‘Gross Domestic Product. Current Prices. JPY, Billions. Annual’. URL: <https://www.e-stat.go.jp/en/stat-search/files?tclass=000001225282&cycle=7&year=20230>. Gross Domestic Product classified by Economic Activities. At current prices. Calendar Year. Gross domestic product.
- Statistics Bureau of Japan (2026*d*), ‘Gross Operating Surplus. Current Prices. JPY, Billions. Annual’. URL: <https://www.e-stat.go.jp/en/stat-search/files?tclass=000001225282&cycle=7&year=20230>. Income and Outlay Accounts classified by Institutional Sectors. Total Economy. Allocation of Primary Income Account. CY. Operating surplus, gross.

- Statistics Bureau of Japan (2026*e*), ‘Net Capital Stock. Chain Volume Measure. JPY, Billions. Annual’. URL: <https://www.e-stat.go.jp/en/stat-search/files?tclass=000001225281&cycle=7&year=20230>. Net Capital Stocks of Fixed Assets classified by Institutional Sectors and Economic Activities. In real terms. Total Fixed Assets.
- Statistics Bureau of Japan (2026*f*), ‘Net Capital Stock. Current Prices. JPY, Billions. Annual’. URL: <https://www.e-stat.go.jp/en/stat-search/files?tclass=000001225281&cycle=7&year=20230>. Net Capital Stocks of Fixed Assets classified by Institutional Sectors and Economic Activities. At current prices. Total Fixed Assets.
- Statistics Canada (2026*a*), ‘Compensation of Employees. Current Prices. CAD, Millions. Annual’. URL: <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3610022101>. Time series key: 62783206.
- Statistics Canada (2026*b*), ‘End-Year Net Stock of Fixed Assets. Chain Volume Measure. CAD, Millions. Quarterly’. URL: <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3410016301>. Time series key: 103844362.
- Statistics Canada (2026*c*), ‘End-Year Net Stock of Fixed Assets. Current Prices. CAD, Millions. Quarterly’. URL: <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3410016301>. Time series key: 103844194.
- Statistics Canada (2026*d*), ‘Gross Domestic Product. Chain Volume Measure. CAD, Millions. Annual’. URL: <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3610069901>. Time series key: 1592185761.
- Statistics Canada (2026*e*), ‘Gross Domestic Product. Current Prices. CAD, Millions. Annual’. URL: <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3610069901>. Time series key: 1592185792.
- Statistics Canada (2026*f*), ‘Gross Operating Surplus. Current Prices. CAD, Millions. Annual’. URL: <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3610022101>. Time series key: 62783209.
- Statistics New Zealand (2026*a*), ‘Compensation of Employees. Current Prices. NZD, Millions. Annual’. URL: <https://www.stats.govt.nz/large-datasets/csv-files-for-download>. National accounts (income and expenditure): Year ended March 2025. GDP Breakdown: Time series key: SNEA.SG03NAC01D10M4.
- Statistics New Zealand (2026*b*), ‘Gross Domestic Product. Chain Volume Measure. NZD, Millions. Annual’. URL: <https://www.stats.govt.nz/large-datasets/csv-files-for-download>. Gross domestic product: September 2025 quarter. Time series key: SNEA.SG00RAC00B15.

- Statistics New Zealand (2026*c*), ‘Gross Domestic Product. Current Prices. NZD, Millions. Annual’. URL: <https://www.stats.govt.nz/large-datasets/csv-files-for-download>. Gross domestic product: September 2025 quarter. Time series key: SNEA.SG00NAC00B15.
- Statistics New Zealand (2026*d*), ‘Gross Operating Surplus and Mixed Income. Current Prices. NZD, Millions. Annual’. URL: <https://www.stats.govt.nz/large-datasets/csv-files-for-download>. National accounts (income and expenditure): Year ended March 2025. GDP Breakdown: Time series key: SNEA.SG03NAC01B02T4.
- Statistics New Zealand (2026*e*), ‘Net Capital Stock. Chain Volume Measure. NZD, Millions. Annual’. URL: <https://www.stats.govt.nz/large-datasets/csv-files-for-download>. National accounts (industry production and investment): Year ended March 2024. Capital stocks. Time series key: SNEA.SG07RAC05K90ZZ99.
- Statistics New Zealand (2026*f*), ‘Net Capital Stock. Current Prices. NZD, Millions. Annual’. URL: <https://www.stats.govt.nz/large-datasets/csv-files-for-download>. National accounts (industry production and investment): Year ended March 2024. Capital stocks. Time series key: SNEA.SG07NAC05K90ZZ99.
- Statistics Norway (2026*a*), ‘Compensation of Employees. Current Prices. NOK, Millions. Annual’. URL: <https://www.ssb.no/en/statbank/table/09170>. Compensation of Employees. Current Prices (NOK million). Total industry.
- Statistics Norway (2026*b*), ‘Consumption of Fixed Capital. Current Prices. NOK, Millions. Annual’. URL: <https://www.ssb.no/en/statbank/table/09170>. Consumption of fixed capital. Current Prices (NOK million). Total industry.
- Statistics Norway (2026*c*), ‘Gross Domestic Product. Chain Volume Measure. NOK, Millions. Annual’. URL: <https://www.ssb.no/en/statbank/table/09189>. Constant 2015 prices (NOK million). Gross domestic product, market values.
- Statistics Norway (2026*d*), ‘Gross Domestic Product. Current Prices. NOK, Millions. Annual’. URL: <https://www.ssb.no/en/statbank/table/09189>. Current prices (NOK million). Gross domestic product, market values.
- Statistics Norway (2026*e*), ‘Net Capital Stock. Chain Volume Measure. NOK, Millions. Annual’. URL: <https://www.ssb.no/en/statbank/table/09181>. Fixed assets. Constant 2015 prices (NOK million). Total industry.
- Statistics Norway (2026*f*), ‘Net Capital Stock. Current Prices. NOK, Millions. Annual’. URL: <https://www.ssb.no/en/statbank/table/09181>. Fixed assets. Current prices (NOK million). Total industry.
- Statistics Norway (2026*g*), ‘Net Operating Surplus and Mixed Income. Current Prices. NOK, Millions. Annual’. URL: <https://www.ssb.no/en/statbank/table/09170>. Operation surplus. Current Prices (NOK million). Total industry.

World Bank (2026), ‘Population, Total. Persons. Annual.’ Retrieved from FRED, Federal Reserve Bank of St. Louis; URL: <https://fred.stlouisfed.org>. Time series key: POPTOTCCA647NWDB, where CC is the two-digit country code in uppercase letters.